

Asymmetric Information

- IT-250: Digital Economy
- Unit 5 – Lecture 2
- Concepts & Determinants

Link with Previous Lecture

- Information helps decisions
- Not everyone has same information
- Unequal information creates problems

Today: Asymmetric Information

What Does Asymmetric Mean?

Asymmetric = Not equal

Example:

- One student knows exam questions
- Other student does not

Information is unequal

Definition

Asymmetric Information occurs when:

- One party in a transaction
- Has more or better information
- Than the other party

Why Is This Important?

- Causes unfair outcomes
- Encourages cheating
- Reduces trust
- Leads to market failure

Daily Life Example

Fruit seller:

- Knows fruit is old

Buyer:

- Thinks fruit is fresh

Buyer pays full price → fruit spoils

Market Example: Used Goods

- Seller knows real condition
- Buyer sees only appearance
- Price decided with incomplete info

Buyer faces higher risk

Why Sellers Know More

- They produced or used the product
- They hide negative information
- They want higher profit

Information advantage = power

Why Buyers Know Less

- Limited time
- Limited technical knowledge
- Depend on seller's words

Information is costly

Two Major Problems

- Asymmetric information causes:
- 1. Adverse Selection
- 2. Moral Hazard

Adverse Selection

- **Adverse Selection happens when:**
- One side knows more **before the deal**
- The other side is unsure
- So **bad options get chosen**
- **Good options disappear**

Example: dishonest sellers dominate

Adverse Selection

Imagine a basket of apples 🍏

- Some apples are fresh
- Some apples are rotten
- Seller knows which are rotten
- Buyer does not

Buyer thinks:

“Some apples might be bad, so I will pay low price.”

What happens?

- People with fresh apples say ❌ “Not worth selling”
- People with rotten apples say ✅ “Good deal!”

End result:

👉 Only rotten apples remain

Adverse Selection

This is a famous economics story.

- **Good cars** = peaches 🍑
- **Bad cars** = lemons 🍋

Buyers can't tell the difference.

So:

- Buyers offer low price
- Good car sellers leave
- Only bad cars stay



Used car market quality falls

Adverse Selection



Simple Cause → Effect Chain

Asymmetric Information



Buyer uncertainty



Low price / strict rules



Good sellers leave



Bad sellers dominate



Market failure

Moral Hazard

- One party becomes careless
- Because someone else bears the risk

Example: insured person acts recklessly

Moral hazard happens when someone becomes careless after the deal is done.

Moral Hazard

Imagine you have insurance 

Before insurance:

- You walk carefully

After insurance:

- “If I fall, insurance will pay!”
- You become careless 

 This is **Moral Hazard**

Moral Hazard - Examples



Bike Insurance Example

- You insure your bike 
- You park it anywhere, even risky places
- “If it’s stolen, insurance will cover it”
- You act less carefully → insurer bears the loss



Bank / Loan Example (Nepal)

- Bank gives a loan to a borrower
- Borrower knows the bank will handle legal action
- Borrower becomes careless with repayment
- Bank bears some risk → moral hazard



Workplace Example

- Employee is covered by a company’s health plan
- Knows mistakes at work won’t be punished immediately
- Takes shortcuts or risks → company suffers

Determinants of Asymmetric Information

- Cost of information
- Access to information
- Product complexity
- Technology
- Regulation

Cost of Information

- Getting correct info costs money & time
- Not everyone can afford it

Poor consumers suffer more

- You want to know if a toy is safe.
- You could pay 500 NPR to test it, or ask a friend.
- Some kids may just guess because testing is expensive → may end up with a broken toy.

Nepal Example:

- Checking whether a manpower agency abroad is genuine requires research, phone calls, maybe even travel.
- Some workers just trust the agency because checking costs too much → risk of bad jobs.

Access to Information

- Internet availability
- Education level
- Language barriers

Access is unequal

- Even if information exists, not everyone can **reach it**.
- Barriers: location, internet, education, language.

Nepal Example:

- In rural Nepal, people may not know market prices for rice or mobile phones.
- Urban buyers can compare online → get better deals.

Product Complexity

Some products are hard to understand:

- Insurance
- Financial products
- Mobile specifications

Sellers exploit complexity

Insurance policies or financial products in Nepal:

- Premium, coverage, hidden clauses...
- Many people can't fully understand → asymmetric information exists.

Technology

- Technology can reduce info gaps
- But can also create new gaps

Platforms know users better than users

- Daraz, SastoDeal, or eSewa use algorithms:
- Show you what you might buy
- Sellers may use ads to mislead → still asymmetric information
- But reviews & ratings help reduce uncertainty.

Regulation & Law

- Weak regulation → more cheating
- Strong regulation → transparency

Government plays key role

- Government can regulate e-commerce:
- Mandatory return policies
- Accurate product descriptions
- Weak enforcement → fake ads & scams flourish

Nepal Context

- Growing digital market
- Weak enforcement in some sectors
- Low digital literacy

Asymmetric info is common

Nepal Example: Manpower Agencies

- Agency knows real job condition
- Worker trusts agent
- Reality different abroad

Classic asymmetric info case

Nepal Example: Education Consultancy

- Consultancy promises scholarships
- Student lacks correct info
- Reality different later

Nepal Example: Online Ads

- Edited photos shown
- Buyer expects high quality
- Delivered product is poor

Impact on Consumers

- Financial loss
- Stress & dissatisfaction
- Reduced market trust

Impact on Businesses

- Honest firms suffer
- Reputation damage
- Unhealthy competition

Impact on Economy

- Reduced market participation
- Inefficient resource allocation
- Slower economic growth

How Can It Be Reduced?

- Transparency
- Reviews & ratings
- Certification
- Regulation
- Technology

Key Exam Keywords

- Asymmetric information
- Information gap
- Adverse selection
- Moral hazard
- Market failure

Simple Summary

- Asymmetric = unequal information
- Sellers often know more
- Causes unfairness
- Very common in Nepal